20 November 1962
SUPERSEDING
MIL-S-22128 (NOrd)
31 July 1959

MILITARY SPECIFICATION

SQUIB, ELECTRIC, MARK 2 MODS

CARTRIDGE, IMPULSE, MARK 45 MOD 0

This specification has been approved by the Bureau of Naval Weapons, Department of the Navy.

1. SCOPE.

1.1 Scope. This specification covers the Squib, Mk 2 Mods referred to herein as the squib.

2. APPLICABLE DOCUMENTS.

2.1 The following documents of the issue in effect on date of invitation for bids form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

PPP-B-621	Boxes, Wood, Nailed and Lock-Corner.

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.

FSC 1340

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MIL-STD-129	Marking for Shipment and Storage.
MIL-STD-350	Jolt Test for Use in Production of Fuzes.
MIL-STD-353	Transportation Vibration Test for Use in Production of Fuzes.
MIL-STD-414	Sampling Procedures and Tables for Inspection by Variables.

DRAWINGS

Bureau of Naval Weapons
(Code Ident 10001)

LD 252961	Shipping (Packing) Condition (For 100 Squibs Mk 1 and 2 Mods).
LD 252968	Preserved (Packing) Condition (For 4 Squibs Mk 1 and 2 Mods).
LD 256003	Ammunition Component Box Mk 2 Mod 0.
LD 267888	List of Drawings, Subassemblies, Parts, and Specifications for Squib, Electric Mk 2 Mods.
SK 378738	Firing Circuit.
SK 378739	Foil Switch Assembly and Details.
SK 378942	Squib Output Test Assembly and Details.
657306	Squib, Mk 2 Mod 0 and 1 Assembly.

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PUBLICATIONS

Bureau of Naval Weapons
(Code Ident 10001)

Technical Memorandum (TM) 1055	Operating Manual and Description of Pyrotechnic Time Sequence Measuring Equipment.

(Copies of TM 1055 may be obtained from Naval Ordnance Test Station, China Lake, California.) (Copies of documents required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

Code of Federal Regulations

49 CFR 71-78	Interstate Commerce Commission Rules and Regulations for the Transportation of Explosives and Other Dangerous Articles.

(The Interstate Commerce Commission regulations are now a part of the Code of Federal Regulations and are available from the Superintendent of Documents, Government Printing Office, Washington 25, D.C. Orders for the above publication should cite "the latest issue and supplements thereto.")

3. REQUIREMENTS.

3.1 Preproduction sample. Unless otherwise specified in the contract or order, there shall be a preproduction sample consisting of 1000 squibs. The preproduction sample shall be manufactured using the methods proposed for production. The preproduction sample shall be delivered to the activity designated in the contract or order (see 6.2) for preproduction and product verification testing (see figure 2) in accordance with this specification. Any production by the supplier prior to the acceptance of the preproduction sample will be at the supplier's risk.

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3.2 Compliance with documents. The squib shall be fabricated and assembled in accordance with LD 267888 including all drawings, specifications and other documents listed thereon and shall meet all requirements of this specification.

3.3 Performance and product characteristics. (See figure 1.)

3.3.1 Electrical continuity and resistance. The electrical circuit of each squib shall be continuous and shall meet the requirements of DWG 657306.

3.3.2 Dielectric characteristics. The resistance between the connected lead wires and the case shall be not less than 50 megohms.

3.3.3 No-fire sensitivity. The squib shall not fire when energized with 1.00 ampere, 24 volts direct current (dc) for a minimum of five seconds.

3.3.4 Firing time. When subjected to 5.0 plus or minus 0.15 amperes dc at plus 70 plus or minus 5 degrees Fahrenheit (F), the squib shall fire in 0.0056 second or less.

3.3.5 Output limits. The squib shall be fired by 5.0 plus or minus 0.15 amperes dc at plus 70 plus or minus 5 degrees F and shall produce a cavity with the capacity of 3 to 15 milliliters in the foam polystyrene block. (See SK 378942.)

3.3.6 Jolt resistance. The squib shall withstand the jolt test of 4.6.1. Thereafter, it shall meet the requirements of 3.3.1 and 3.3.2 and shall function in accordance with 4.5.4.

3.3.7 Vibration resistance. The squib shall withstand the vibration test of 4.6.2. Thereafter, it shall meet the requirements of 3.3.1 and 3.3.2 and shall function in accordance with 4.5.4.

3.3.8 Watertightness. The squib shall be watertight and shall show no evidence of leakage when tested in accordance with 4.6.3 and thereafter shall function in accordance with 4.5.4.

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3.3.9 Workmanship. The squib exterior shall be free from flux, oil, grease, explosive, foreign material and shall be free from visible blemishes.

4. QUALITY ASSURANCE PROVISIONS.

4.1 The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to assure that supplies and services conform to the prescribed requirements.

4.1.1 Acceptance. Authority and responsibility for acceptance shall rest with the cognizant Government representative.

4.2 Sampling. Except as otherwise specified herein, the following documents shall govern the establishment of sampling plans, and procedures for inspection, acceptance and rejection.

(a) Standard MIL-STD-105 for attributes.

(b) Standard MIL-STD-414 for variables.

4.3 Preproduction sample. The preproduction sample shall conform to requirements of 3.1. Failure to meet these requirements shall be cause for rejection of the preproduction sample and shall also be cause for cancellation of the contract or order.

4.4 Lot. Unless otherwise specified in the contract or order, the lot size shall be 10,000 squibs (see 6.2).

4.5 Acceptance tests. (See figure 2.)

4.5.1 Electrical continuity and resistance test. Every squib shall be subjected to an electrical continuity test using a resistance testing current of less than 10 milliamperes dc. Each squib that does not meet the requirement of 3.3.1 shall be rejected.

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4.5.2 Dielectric test. A dielectric test shall be conducted for verification of conformance to the requirement of 3.3.2. The test potential shall be either 1000 plus or minus 100 volts, 60 cycle alternating current (ac) or 500 plus or minus 50 volts dc. The test potential shall be applied for not less than one second. Sampling shall be in accordance with Standard MIL-STD-105, Acceptable Quality Level (AQL) 1.0 percent defective.

4.5.3 No-fire sensitivity test. The squib shall be tested to determine compliance with requirements of 3.3.3. Sampling shall be in accordance with Standard MIL-STD-105, AQL 0.065 percent defective. The samples used for this test shall not be delivered to the Government as part of production.

4.5.4 Firing time test. The firing time test shall be conducted in accordance with figures 1, 2, 6, and 13, Technical Memorandum (TM) 1055, and BuOrd DWGS SK 378738, and SK 378739. A sample shall be selected in accordance with Standard MIL-STD-414, Variability Unknown, AQL 1.0 percent defective. In addition, failure to meet the requirements of 3.3.4 or occurrence of one or more duds or misfires shall be cause for rejection of the lot represented.

4.5.5 Output limits test. The squib shall meet the requirements of 3.3.5. Test equipment shall be in accordance with SK 378942. A sample shall be selected in accordance with Standard MIL-STD-414, Variability Unknown, AQL 1.0 percent defective. In addition, if one or more squibs fail to fire, the lot represented shall be rejected.

4.6 Preproduction and product verification tests. These tests shall be conducted on the preproduction sample (see figure 2), and on each rejected lot resubmitted for acceptance. In addition, the Government reserves the right any time during production to test random samples of squibs for product verification. (See figure 2.)

4.6.1 Jolt test. The squib shall be subjected to the jolt test in accordance with Standard MIL-STD-350 and thereafter shall meet the requirements of 3.3.6.

4.6.2 Vibration test. The vibration test shall be conducted in accordance with Standard MIL-STD-353. The squib shall be secured in a suitable rigid-mounting fixture, and the leads taped to it. Thereafter, the squib shall meet the requirements of 3.3.7.

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4.6.3 Watertightness test. The squibs shall be subjected to a cycle of four pressure conditions for a period of three to five minutes each. The cycle shall be repeated 21 times continuously. The conditions (a), (c), (d), (f) and procedures (b) and (e) are as follows:

(a) Submerge squibs in plus 160 plus or minus 10 degrees F water at absolute pressure of approximately 58 inches of mercury.

(b) Drain or pump water from chamber.

(c) Reduce chamber absolute pressure to not more than 5 inches of mercury.

(d) Fill chamber with tap water to an absolute pressure of approximately 58 inches of mercury. Temperature differential between tap water and hot water, condition (a), shall be not more than 90 degrees F.

(e) Drain or pump water from chamber.

(f) Reduce chamber absolute pressure to not more than 5 inches of mercury.

Thereafter, the squibs shall function in accordance with 4.5.4. Failure to meet the requirements of 3.3.8 or occurrence of one or more duds or misfires shall be cause for rejection of the lot.

4.7 Visual examination. The squib shall be visually inspected for workmanship, and conformance to preservation, packaging, packing and marking requirements of this specification.

5. PREPARATION FOR DELIVERY. (See 6.2.)

5.1 Preservation and packaging.

5.1.1 Level A. In addition to detailed requirements herein and in applicable referenced documents, Level A preservation and packaging shall be in accordance with the Code of Federal Regulations 49 CFR 71-78.

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5.1.1.1 Preservation. Preserved packaging of four squibs shall be in accordance with LD 252968.

5.1.1.2 Unit packaging. The shipping packaging condition for 100 squibs shall be in accordance with LD 252961.

5.2 Packing.

5.2.1 Level A. In addition to detailed requirements herein and in applicable referenced documents, Level A packing shall be in accordance with the Code of Federal Regulations 49 CFR 71-78.

5.2.2 Packing. One hundred boxes, each containing 100 squibs (for a total of 10,000 squibs), packaged in accordance with 5.1.1.2 shall be packed in an Ammunition Components Box Mk 2 Mod 0, LD 256003, or in a nailed wooden box conforming to the requirements of Specification PPP-B-621.

5.3 Marking for shipment and storage.

5.3.1 Standard marking. Unless otherwise specified in applicable documents, standard marking shall be in accordance with Standard MIL-STD-129.

5.3.2 Special marking. Special marking shall conform to the requirements of the Code of Federal Regulations 49 CFR 71-78.

6. NOTES.

6.1 Intended use. The squib is intended as a part of the ignition system for rocket motor propellant and any other use deemed appropriate and applicable.

6.2 Ordering data. Procurement documents should specify the following:

(a) Number and date of this specification.

(b) Preproduction lot if different from 3.1.

(c) Name and location of testing activity designated to evaluate the preproduction lot 3.1.

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(d) Production lot if different from 4.4.

(e) Mod 0 or Mod 1 required.

(f) Invocation of Specification MIL-Q-9858.

6.3 Preservation, packaging and packing levels. For definition of preservation, packaging and packing levels refer to Standard Fed Std No. 102.

6.4 General safety precautions. The loading, assembly and handling of the item covered by this specification, and the subassemblies thereof, involve hazardous operations and therefore require suitable explosives safety precautions. Use of this specification will not be construed as to relieve the supplier or manufacturer of responsibility for the safety of his operations. Listed below are certain minimum provisions which a supplier or manufacturer (which explosively loads the item covered) should observe in order to fulfill his responsibility for safety. At Bureau of Naval Weapons, Navy Department, and other Government plants, these provisions are mandatory. Such other warnings and precautions, pertinent to the operational effectiveness or safety during use or loading of the specified item, are included in the detail technical requirements of the specification.

6.4.1 All loading operations should be conducted in a neat and orderly manner.

6.4.2 Safe equipment and methods should be utilized for transporting and handling explosives and loaded parts. Where required, remote control barricaded handling equipment shall be used for explosives operations, such as mixing, pouring, weighing, charging, sifting, drying, pressing, casting, crimping, etc.

6.4.3 Personnel handling detonators, primers, delay elements, lead-ins, boosters, and related parts which affect functioning, should insofar as practicable, avoid using bare fingers or improper equipment in order to prevent damage, corrosion, or deterioration from perspiration or other contaminating deposits.

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6.4.4 The exposure of explosive materials and related parts should be so controlled as to minimize the absorption of moisture from the atmosphere or other sources during loading and handling operations.

6.4.5 All explosives and completely or partially loaded items should be stored in suitable storage magazines located in accordance with the American Table of Distances (ATD) or other applicable safety standards; and, while in process, in safety lockers and chests if in loading rooms, or in adequate ready or service magazines located in accordance with intraplant distances when outside of loading rooms. For Navy managed explosives loading plants, the provisions of the Armed Services Explosives Safety Board covering quantity-distance relations for explosives will apply.

6.4.6 Proper care must be exercised at all times to protect personnel from accidents, fires or explosions, and to limit damage to equipment and loading areas. In this connection, the precautionary measures in the following paragraphs should be observed.

6.4.6.1 Employ properly proportioned and properly located protective barricades, screens or shields at all required points.

6.4.6.2 Keep only minimum limited quantities of explosives and completed or partially loaded parts present at each stage of operation.

6.4.6.3 Keep explosives and explosive parts in approved covered receptacles with covers in place when material is not being taken out of or put into the receptacles. Where necessary, receptacles should be conductive to ground electrostatic charges.

6.4.6.4 Protect operations from electrostatic charges by effectively grounding all machinery, equipment, and fixtures; and, where necessary, employ suitable grounded conductive coverings for floors, work benches and tables, and workers' conductive shoes. Workers' clothing of a type to minimize the accumulation of static charges should be employed. Fabrics such as silk and nylon, which promote static generation should

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be avoided. Additional grounding devices such as grounded bracelets for workers should be employed where operations are conducted with items which are unusually sensitive to initiation by static electricity. Such items include initiating explosives, tracer mixtures, and low-energy type electric primers, detonators and squibs. The latter types of items should have the free ends of lead wires bared and twisted together, and be packed in relatively small groups wrapped in bare non-insulated aluminum foil or other uncoated metal foil. During assembly and processing operations such sensitive electric items should be short circuited by clips or other devices until installed with safety shunt in the final device. Additional precautions for these items should include mechanical shielding to contain or deflect fragments and blast, also electrical shielding of these items from induced electric currents generated by sources such as lightning, static, radiations from communications apparatus, radar, or high frequency heat apparatus, etc. Where necessary for safety, humidity of work rooms should be appropriately increased, as required to lessen electrostatic effects but without inducing excessive moisture absorption by any of the components of the item being loaded.

6.4.6.5 Protect all explosive operations from effects of electric current originating from equipment such as soldering irons, heaters, switches, wiring, motors, lights, test instruments, etc., by suitable insulation, grounding separation or shielding. Such electric sources may initiate explosives by heat, sparks, arcs, or due to completing an electric circuit through an electric primer, detonator, or squib. Circuits may be inadvertently completed, for example, from a defective electric soldering iron through a grounded contact. All electric type primers, detonators or squibs provided with wire leads should have the free ends of the wires bared and twisted together to short circuit each unit, except when in process of assembly into a finished item. Where practicable, removable short circuiting clips, or other devices should be employed during manufacturing operations involving electric primers, detonators or squibs.

6.4.6.6 Enforce, where necessary, the wearing of suitable safety footwear, gloves, goggles, respirators, and impregnated garments to protect personnel against burns, poisoning and associated industrial hazards.

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6.4.6.7 Allow no fires or exposed electrical or other sparking equipment, and little or no flammable material to be present in loading, handling and storage spaces. Enforce proper "Match" and "No Smoking" rules where necessary.

6.4.6.8 Enforce good housekeeping and maintain effective policing, inspection and supervisory methods throughout the loading area and surroundings. Employ effective cleaning methods periodically to minimize the accumulation of explosives and explosive dust and other contamination upon, and assure its removal from floors, walls, ceilings, ledges, tables, benches, piping, and equipment or the items loaded; also, clean up any spilled material immediately.

Notice. When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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	Test	Lower Limit	Upper Limit	AQL %
4.5.1	Electrical continuity and resistance test	Dwg 657306		---
4.5.2	Dielectric test -megohm	50	---	1.0
4.5.3	No-fire sensitivity test	---	1.0 amp. 24 v	
		5 sec	---	0.065
4.5.4	Firing time test -second	---	.0056	1.0
4.5.5	Output limits test -milliliter	3	15	1.0
4.6.1	Jolt test Firing time -second	---	.0056	---
4.6.2	Vibration test Firing time -second	---	.0056	---
4.6.3	Watertightness test Firing time -second	---	.0056	---

Figure 1. Summary of Required Values and AQLs

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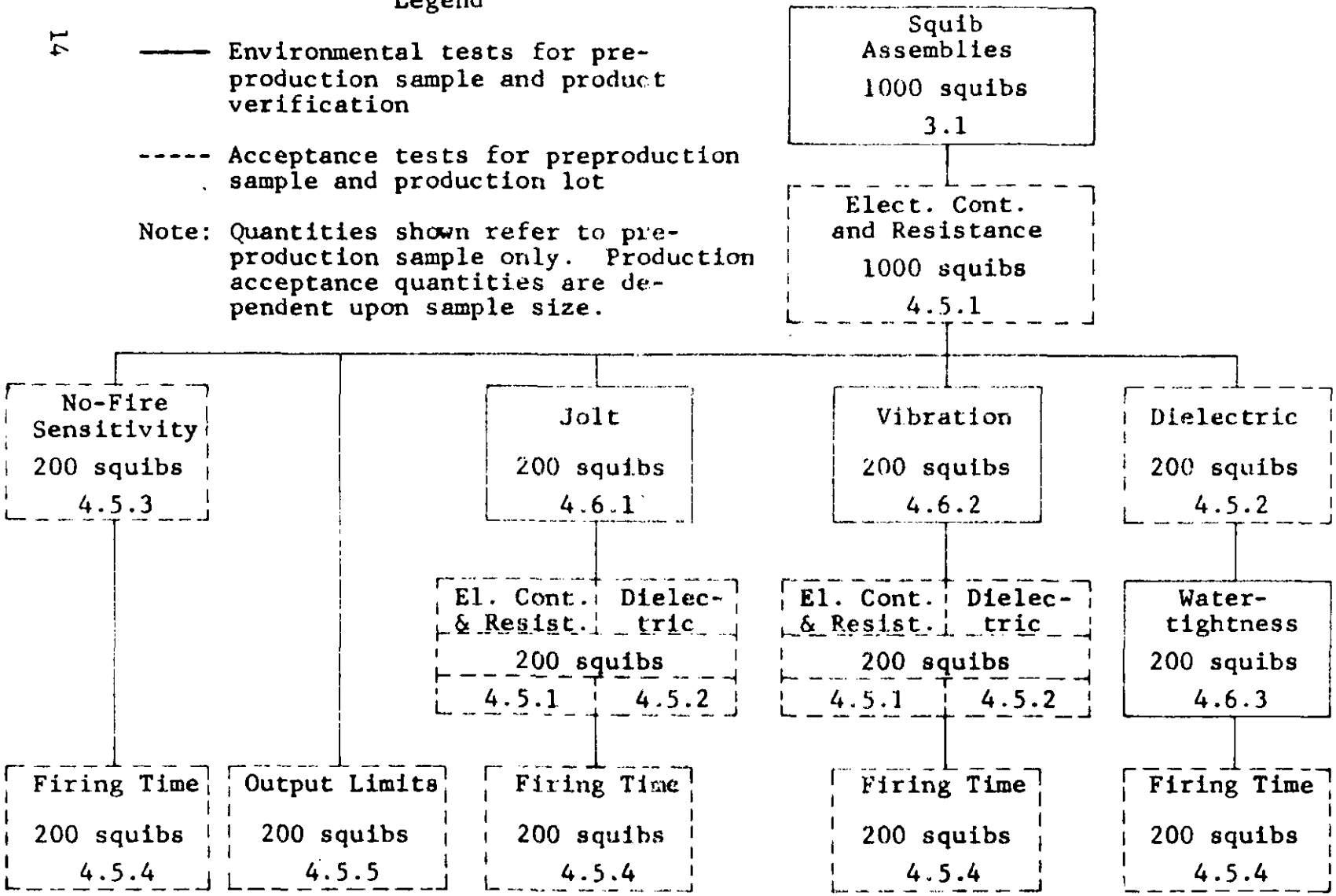


Figure 2. Test Sequence.

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SPECIFICATION ANALYSIS SHEET
NAVWEPS FORM 4121/3 (8-62)

FORM APPROVED
BUDGET BUREAU NO. 45-R309

INSTRUCTIONS

This sheet is to be filled out by Government or contractor personnel involved in the use of this specification in procurement of products for ultimate use by the activity shown on the reverse of this sheet.

This sheet is provided for obtaining information which will aid the activity shown in insuring that through use of this specification suitable products can be procured with a minimum amount of delay and at the least cost.

Comments and the return of this sheet will be appreciated.

Fold on dotted lines on reverse side, staple in corner, and send to the activity shown.

SPECIFICATION *(No. and abbrev. title)*

ORGANIZATION	CITY	STATE

CONTRACT NUMBER	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A DIRECT GOVERNMENT CONTRACT ☐ OR A SUBCONTRACT ☐

1. HAS ANY PART OF THIS SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
(If so, give paragraph number and wording, and recommendations for correcting the deficiencies.)

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?
☐ YES ☐ NO IF YES, IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If additional papers are attached, send this form and papers in an envelope. This form is addressed for use in window envelope when appropriate.)*

SUBMITTED BY *(Print name and title)*	DATE

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